

Vehicle Vitals Solution Storyboards

Last updated: May 20, 2026 Purpose: Convert business and technical requirements into storyboard visuals for beta-test redesign workshops.

1) How To Use This Page

- 1. Start with the Coverage Matrix to confirm requirement coverage.
- 2. For each storyboard, open both linked visuals (Web and Mobile) on this same page.
- 3. Use the Discussion Prompts under each storyboard to decide what to rework.
- 4. Capture decisions against the linked requirement source docs.

2) Coverage Matrix (Business + Technical)

Requirement Stream	Primary Source Docs	Storyboards
Personas, market, business outcomes, GTM	BUSINESS_REQUIREMENTS.md, PROJECT_PLAN.md	SB-01, SB-12
Scope and tier delivery status	REQUIREMENTS.md, RELEASE_SCOPE_MATRIX.md	SB-02, SB-08, SB-12
Product UX flows and IA	PRODUCT_DESIGN.md	SB-02 to SB-07
Architecture and platform boundaries	ARCHITECTURE.md	SB-09
API contracts and Firestore model expectations	API_DATA_MODELS.md	SB-03, SB-04, SB-05, SB-09
Security, auth, access control, data protection	SECURITY_IMPLEMENTATION.md	SB-02, SB-10
Monetization, subscriptions, ads, upgrade triggers	MONETIZATION_STRATEGY.md, PHASE1_IMPLEMENTATION_STATUS.md	SB-08
Near-term release gates and execution sequencing	NEXT_FEATURES_EXECUTION_PLAN.md, R1_COMPLETION_CHECKLIST.md	SB-11, SB-12
Enterprise/org governance and compliance workflows	REQUIREMENTS.md (Enterprise Foundation Update)	SB-10

3) Cross-Platform Story Map

**SB-01 Vision and Persona
Fit**



SB-02 Onboarding and Trust



**SB-03 Vehicle Profile and
VIN**



SB-04 Maintenance Logging



SB-05 Alerts and Calendar



SB-06 Timeline and Insights



SB-07 Export and Data
Portability



SB-08 Monetization and
Upgrade



SB-09 API and Reliability

Layer

```
graph TD; A[Layer] --> B[SB-10 Security and Enterprise Controls]; B --> C[SB-11 Release Gate Validation]; C --> D[SB-12 Growth, KPI, and Roadmap Decisions];
```

SB-10 Security and
Enterprise Controls

SB-11 Release Gate
Validation

SB-12 Growth, KPI, and
Roadmap Decisions

4) Storyboards

SB-01 Vision and Persona Fit

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Responsible owner confidence and reduced maintenance misses
- Fleet and multi-vehicle visibility
- Young driver clarity and approachable UX

Technical requirement focus:

- Cross-platform parity framing (web + mobile)
- Shared capability definitions before UI divergence

Discussion prompts:

- Which persona is underserved in the first 30 seconds?
- Do web and mobile communicate the same value promise?

SB-01-Web



Ads are disabled for this environment

VEHICLE VITALS PLATFORM

Vehicle ownership tools explained with real screens and guided demos

See what the product does today across garage setup, maintenance planning, ownership history, nearby providers, and cross-device access before you sign in.

Create your account

View getting started workflow

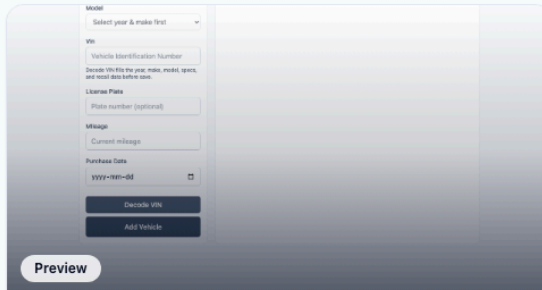
Real workflow coverage

Real application screenshots

Guided product walkthroughs

Flagship capability demos

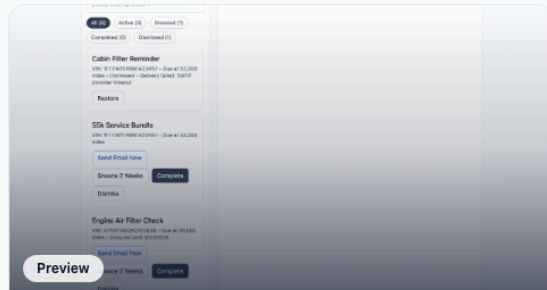
Focused previews that show what each capability is for and where it leads inside the secured app.



VIN Decode & Quick Add

Start with a VIN and move from lookup to complete garage entry in moments.

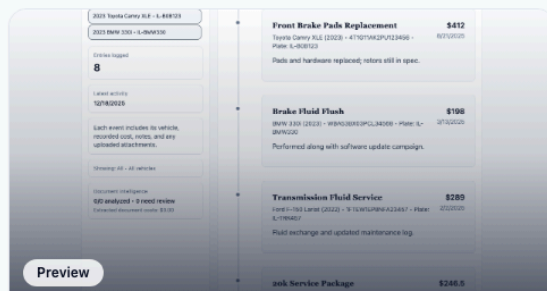
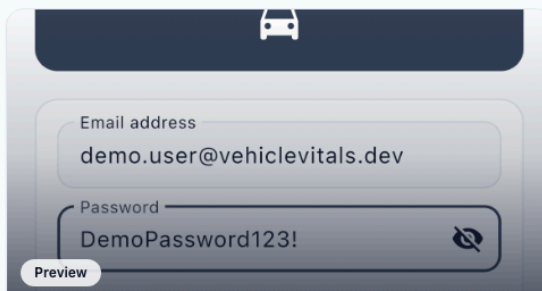
View VIN Demo



Maintenance Planning

See upcoming tasks, completed services, and ownership costs in one timeline.

View Planning Demo



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SB-01-Mobile

15:39



Maps



Calendar



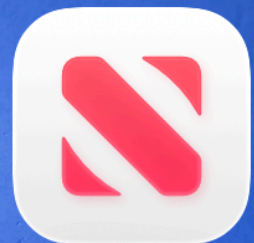
Calendar



Photos



Reminders



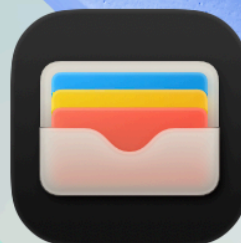
News



Maps



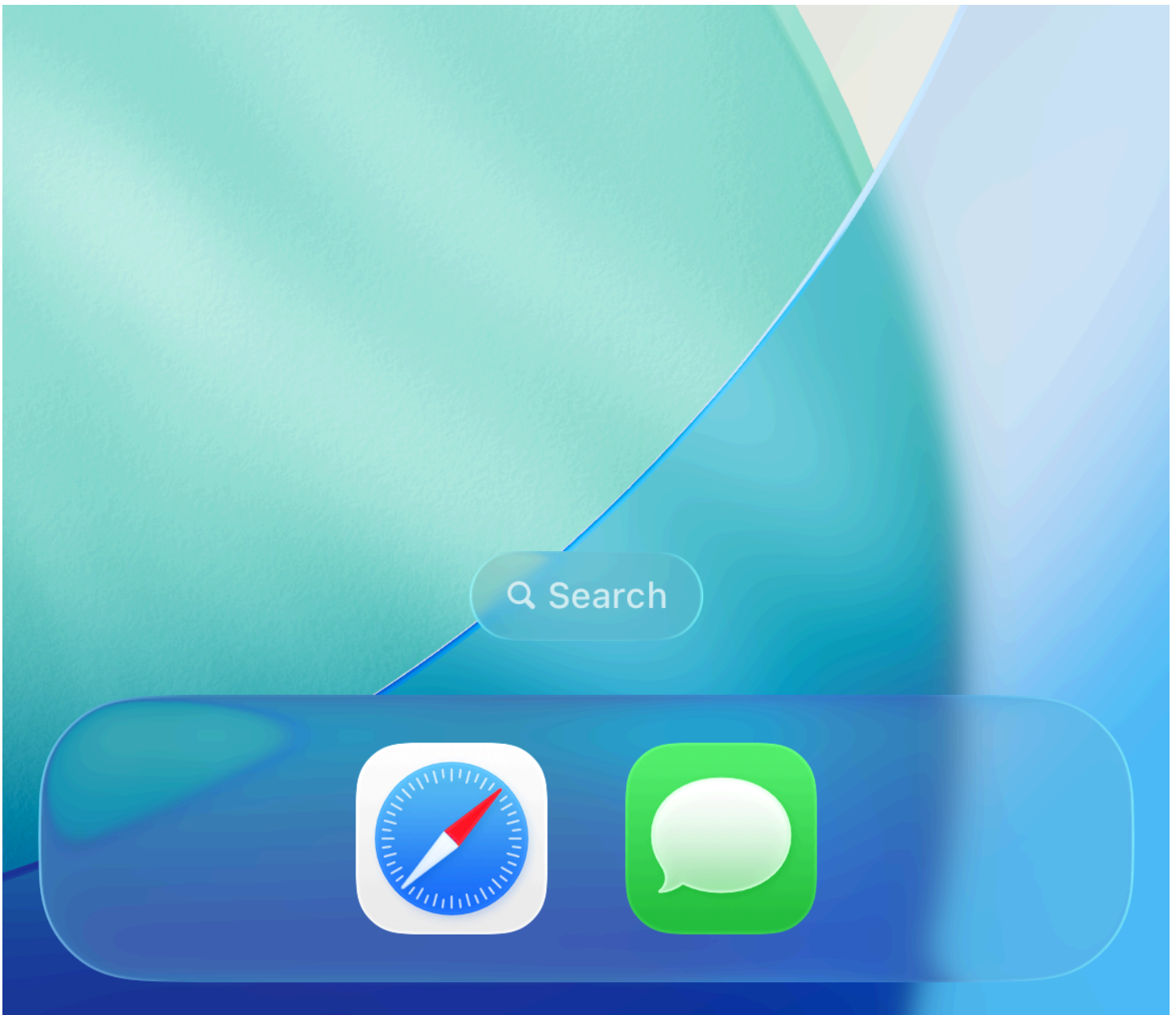
Health



Wallet



Settings



SB-02 Onboarding, Authentication, and Trust

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Fast activation and conversion to first vehicle
- Minimize onboarding drop-off

Technical requirement focus:

- Secure auth, session handling, protected routes
- Policy-aligned sign-up/reset flows

Discussion prompts:

- Where do users hesitate during sign-in/sign-up?
- Which trust signals are missing for first-time users?

SB-02-Web



Need help? [Visit Help](#)

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Sign in to Vehicle Vitals

Access your secure garage, timeline, and upcoming maintenance.

Email address

Password

Show

Sign In

Continue with Google

Continue with Apple

Forgot your password? [Reset it](#)

Don't have an account? [Sign up](#)

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SB-02-Mobile

DEBUG

AdMob has a YouTube channel
Tap for tutorials, screencasts, & more.

Test mode



Vehicle Vitals

Marketing: what Vehicle Vitals delivers

Track maintenance, timeline events, and ownership history across web and mobile.

- VIN-first onboarding and quick setup
- Service timeline and upcoming tasks
- Secure account with synced data
- Cross-platform access and continuity

Sponsored

General RV



GENERAL RV
**Shop the
vans**

Test mode

Shop Now

Next: authenticate to access your secure garage.

Sign In

Create Account

SB-03 Vehicle Setup and VIN Intelligence

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Reduce manual setup effort
- Improve data accuracy for maintenance planning and resale value

Technical requirement focus:

- VIN decoding service contract
- Firestore vehicle schema fidelity and validation

Discussion prompts:

- Which fields should be mandatory vs inferred?
- How should failures in VIN decode degrade gracefully?

SB-03-Web



Ads are disabled for this environment

Add Vehicle

[Back](#)

Enter the VIN, decode it for faster setup, then review or complete the vehicle details before saving.

Vehicle Setup

Year

Select Year

Make

Select Make

Model

Select year & make first

Vin

Vehicle Identification Number

VIN is required to save a vehicle. Decode fills year, make, model, specs, and recall data for you, and you can edit any missing fields before saving.

License Plate

Plate number (optional)

Mileage

Current mileage

Purchase Date

yyyy-mm-dd



Decode VIN

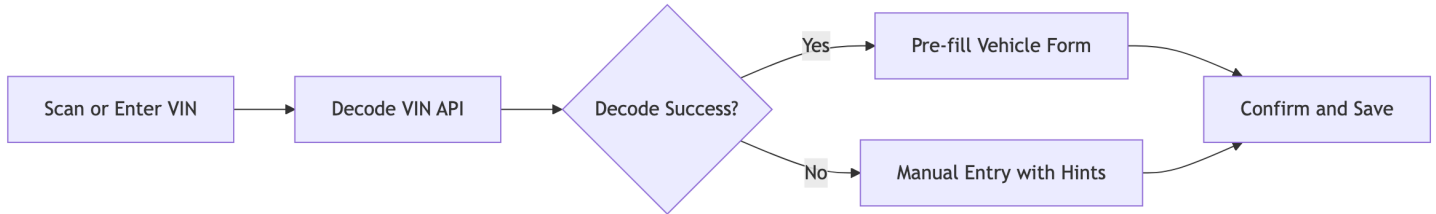
Add Vehicle

Decode is optional, but a VIN is still required as the vehicle's primary identifier.

Decode Preview

Decode a VIN to preview the vehicle profile, recall count, and spec fields that will be saved with this vehicle. If lookup is incomplete, update the form manually and continue.

SB-03-Mobile



SB-04 Maintenance Logging and Record Integrity

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Keep complete history for reliability, warranty, and resale
- Make logging fast enough for routine use

Technical requirement focus:

- Maintenance subcollection structure
- CRUD consistency and validation rules across clients

Discussion prompts:

- Is the logging flow one-screen or multi-step for best completion rate?
- Which fields are required for downstream analytics and reminders?

SB-04-Web



Ads are disabled for this environment

Edit Vehicle

Back

2022 Ford F-150 Lariat • 1FTEW1EP8NFA23457

Vehicle Details

Year

2022

Make

Current: Ford

Model

Current: F-150 Lariat

VIN

1FTEW1EP8NFA23457

Enter VIN and click Decode to prefill vehicle details

License Plate

IL-TRK457

Mileage

51745

Purchase Date

2022-03-04

Decode VIN

Save Changes

Delete Vehicle

Vehicle Insights

Specifications

Body Class

Pickup

Fuel Type

Gasoline

Trim

SuperCrew

Vehicle Type

TRUCK

Drive Type

4WD/4-Wheel Drive/4x4

Active Recalls

Source: NHTSA (temporarily unavailable)

No active recalls found for this vehicle.

Last updated: 5/19/2026

Maintenance

Recommended Maintenance

Calendar Google

Calendar sync is available on Pro and Premium plans.

Ford F-150 Lariat (2022) • Current mileage: 51745

No manufacturer schedules available for this vehicle.

Maintenance History

Export CSV (Pro) Export PDF (Pro)

- Oil + Filter Service

\$118.25

Synthetic blend and OEM filter.

2024-04-15
- Battery Replacement

\$236.75

Cold-cranking amps below threshold during winter test.

2025-12-19

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SB-04-Mobile

Open Vehicle



Tap Add Maintenance



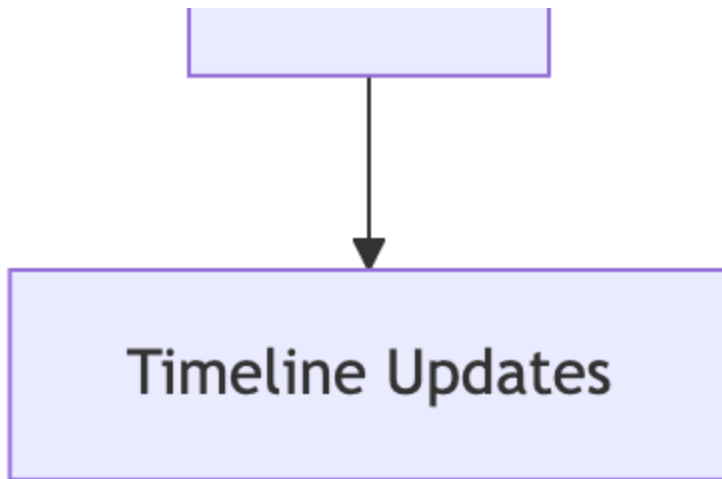
Service Type, Date,
Mileage, Cost



Attach Receipt Optional



Save



SB-05 Alerts, Upcoming Tasks, and Calendar Actions

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Prevent missed maintenance
- Convert reminders into completed service actions

Technical requirement focus:

- Reminder lifecycle actions (snooze, dismiss, complete, reopen)
- Calendar provider flow and callable/HTTP fallback resilience

Discussion prompts:

- Are alert priorities understandable at a glance?
- What should happen when calendar sync fails?

SB-05-Web



Ads are disabled for this environment

Upcoming Tasks

Stay ahead of maintenance with upcoming service recommendations for all your vehicles.

Reminder Center

This page estimates when service should surface based on your lead time and average daily driving. You can still reveal everything and save a reminder early when you want more manual control.

Alerts: **Enabled**
Lead time: **14 days**
Average driving: **35 miles/day**
Current reminder window: **490 miles**

Tasks due within about **14 days at 35 miles/day** appear by default.

[Show all recommendations](#)

[Adjust reminder preferences](#)

Send reminder email to

you@example.com

Use this for one-time reminder emails when you choose Send Email Now. It does not start an automatic email campaign.

All (5) Active (3) Snoozed (1)

Completed (0) Dismissed (1)

See every saved reminder across active, snoozed, completed, and dismissed states.

Cabin Filter Reminder

VIN: 1FTEW1EP8NFA23457 • Due at 52,000 miles • Dismissed • Delivery failed: SMTP provider timeout

[Restore](#)

55k Service Bundle

VIN: 1FTEW1EP8NFA23457 • Due at 55,000 miles

[Send Email Now](#)

[Snooze 14 Days](#)

[Complete](#)

[Dismiss](#)

Engine Air Filter Check

VIN: 4T1G11AK2PU123456 • Due at 39,500 miles • Snoozed until 3/31/2026

[Send Email Now](#)

[Snooze 14 Days](#)

[Complete](#)

[Dismiss](#)

40k Mile Service Due Soon

IN RANGE NOW

0

SAVED REMINDERS

5

OUTSIDE WINDOW

0



All caught up!

No upcoming maintenance tasks fall inside your current reminder window.

[Show all recommendations](#)

[Edit reminder preferences](#)

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SB-05-Mobile

Authenticated Mobile
Session



Open Upcoming Tasks



Load Reminder Queue



Complete, Snooze, or
Dismiss



Persist Reminder Lifecycle State

SB-06 Timeline, Dashboard, and Insight Comprehension

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Help users answer: what happened, what is next, what will it cost?
- Support multi-vehicle oversight and decision confidence

Technical requirement focus:

- Query/filter consistency
- Performance on high history volumes

Discussion prompts:

- Is trend visibility clear enough for action?
- Which metrics are noise vs must-have?

SB-06-Web



Ads are disabled for this environment

Maintenance Timeline

Review completed work across the garage in reverse chronological order.

Timeline Summary

Time Range

All Past Future

Vehicles

All Vehicles Clear

- 2022 Ford F-150 Lariat • IL-TRK457
- 2023 Toyota Camry XLE • IL-B0B123
- 2023 BMW 330i • IL-BMW330

Entries logged

8

Latest activity

12/18/2025

Each event includes its vehicle, recorded cost, notes, and any uploaded attachments.

Showing: All • All vehicles

Document intelligence
0/0 analyzed • 0 need review
Extracted document costs: \$0.00

Battery Replacement**\$236.75**

Ford F-150 Lariat (2022) • 1FTEW1EP8NFA23457 • 12/18/2025
Plate: IL-TRK457

Cold-cranking amps below threshold during winter test.

Front Brake Pads Replacement**\$412**

Toyota Camry XLE (2023) • 4T1G11AK2PU123456 • 8/21/2025
Plate: IL-B0B123

Pads and hardware replaced; rotors still in spec.

Brake Fluid Flush**\$198**

BMW 330i (2023) • WBA53BX03PCL34568 • Plate: IL-BMW330 3/13/2025

Performed along with software update campaign.

Transmission Fluid Service**\$289**

Ford F-150 Lariat (2022) • 1FTEW1EP8NFA23457 • Plate: IL-TRK457 2/2/2025

Fluid exchange and updated maintenance log.

20k Service Package**\$246.5**

Toyota Camry XLE (2023) • 4T1G11AK2PU123456 • Plate: IL-B0B123 11/5/2024

Cabin filter replaced, brake fluid checked.

Oil + Filter Service**\$118.25**

Ford F-150 Lariat (2022) • 1FTEW1EP8NFA23457 • Plate: IL-TRK457 4/14/2024

Synthetic blend and OEM filter.

10k Service and Tire Rotation**\$129.95**

Toyota Camry XLE (2023) • 4T1G11AK2PU123456 • 2/9/2024

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SB-06-Mobile

Authenticated Home
Screen



Vehicle Overview Cards



Timeline Snapshot



Upcoming Alerts Preview



Navigate to Full Timeline

or Upcoming

SB-07 Service Providers and Operational Next Step

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Convert insight into service action
- Improve repair-shop and dealership discoverability

Technical requirement focus:

- Provider lookup callable contracts
- Preference persistence (radius/type/make matching)

Discussion prompts:

- Is provider confidence explained (distance, relevance, fit)?
- Should provider action start inside app or deep-link out?

SB-07-Web



Ads are disabled for this environment

Service Providers

Find nearby repair shops and dealerships using your saved address, radius, and provider preferences.

Search Preferences

Save your home-area search settings here, then rerun the lookup any time you want a fresh nearby-provider list.

Street

City

State

Radius (miles)

25

Provider Type

All



Vehicle Make

BMW



☒ Prioritize my saved vehicle make for dealership results

Find Nearby Providers

Query: US

Results

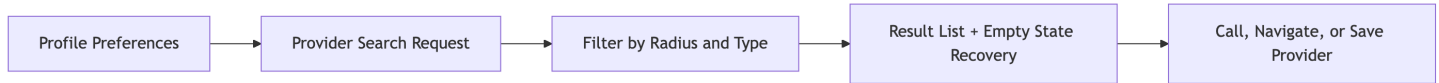
Source: unknown

No providers yet. Run a search to view local options.

Ads are disabled for this environment

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SB-07-Mobile



SB-08 Monetization, Feature Gating, and Upgrade UX

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Sustainable revenue via free/pro/premium paths
- Clear value exchange for upgrades

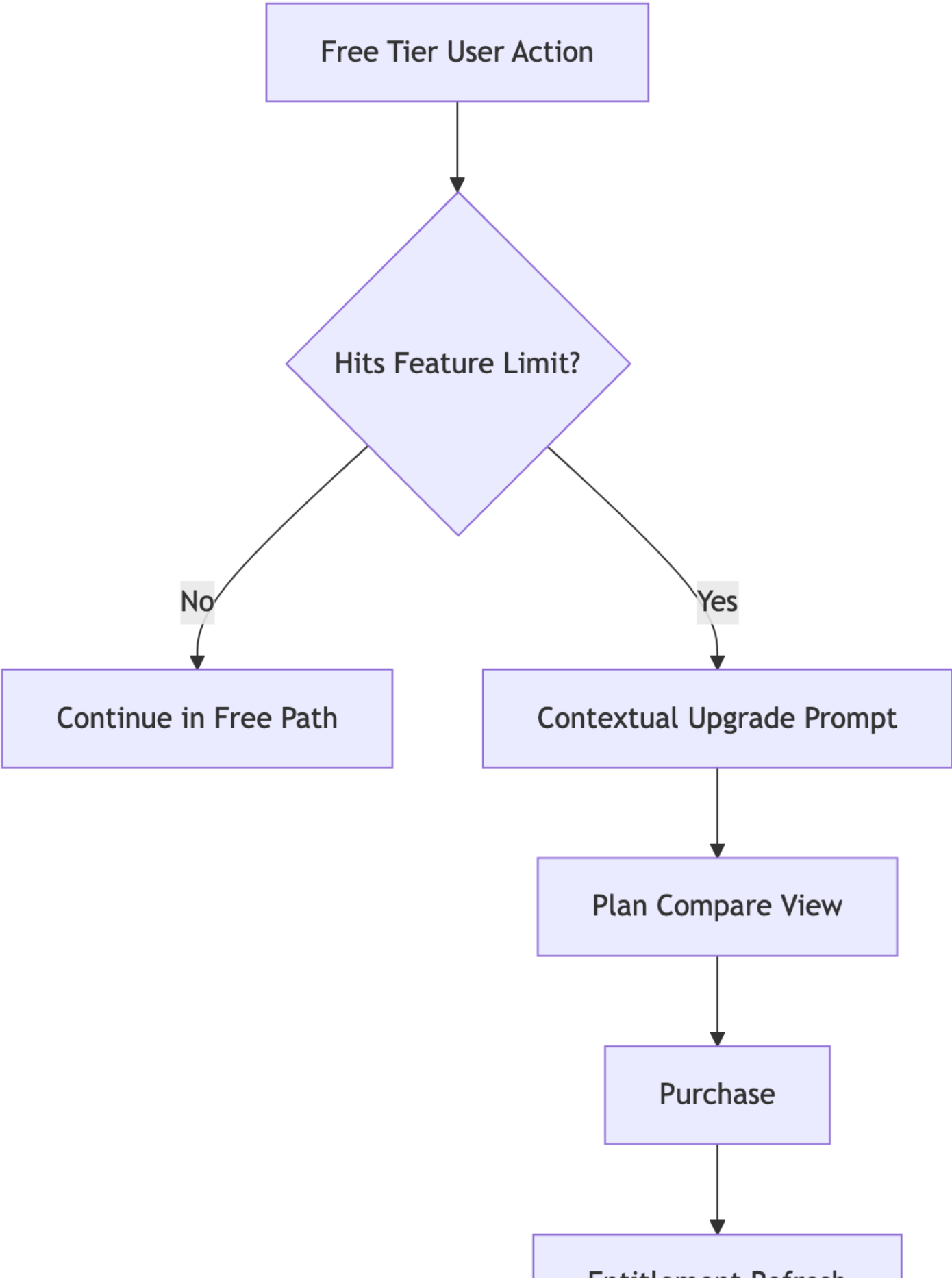
Technical requirement focus:

- Tier entitlement resolution and enforcement
- Ad placement rules and premium ad suppression
- Purchase verification and entitlement sync

Discussion prompts:

- Which upgrade trigger moments feel helpful vs disruptive?
- Do entitlement changes reflect immediately and consistently?

SB-08-Web



Entitlement Refresh



Unlocked Feature + Ad
Reduction

SB-08-Mobile

In-App Paywall



Store Purchase



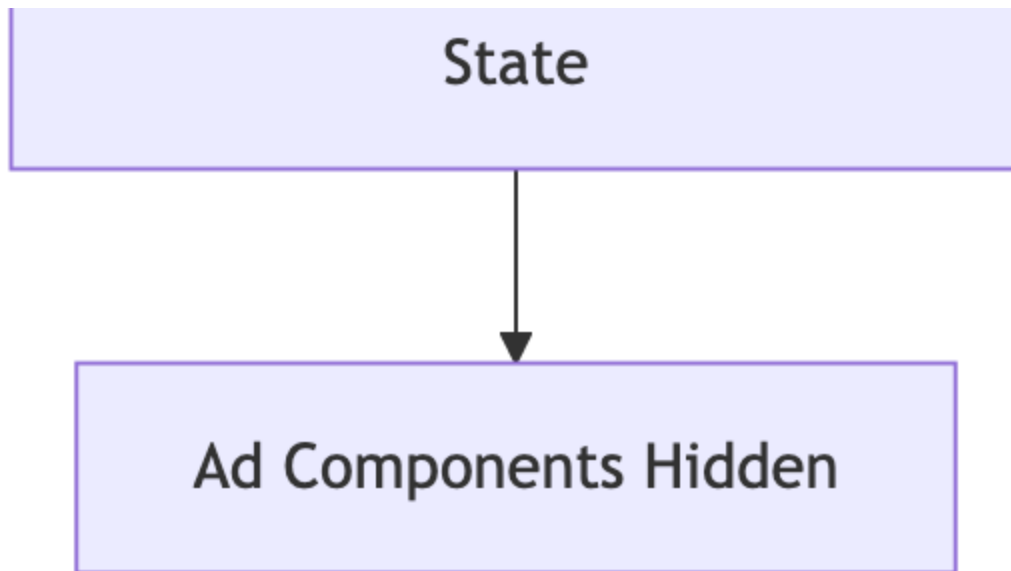
Receipt Verification
Callable



Server Entitlement



Client Refreshes Premium



SB-09 Technical Backbone: Data, APIs, and Offline Behavior

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Reliability and speed protect retention and trust

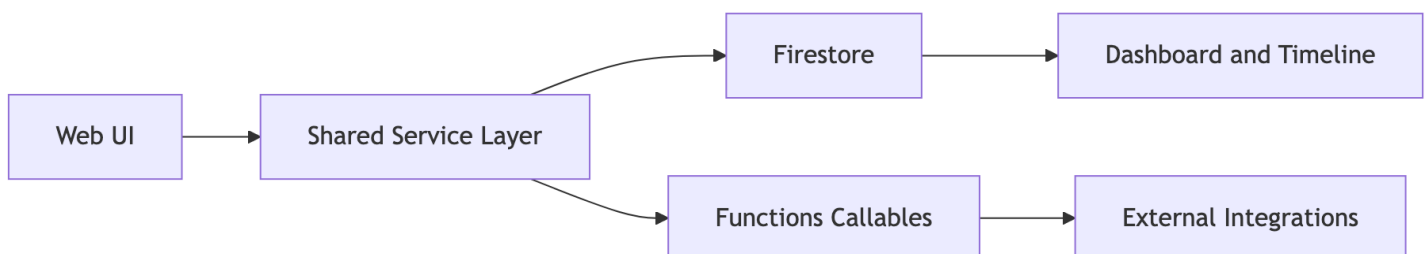
Technical requirement focus:

- Firebase-first boundaries (auth, firestore, functions)
- Shared package contracts
- Caching, retries, and offline behavior

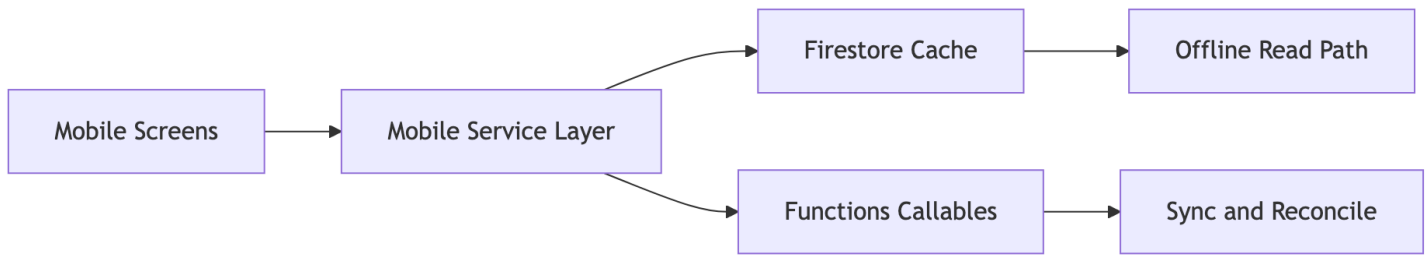
Discussion prompts:

- Which errors should block user action vs queue for retry?
- Are API failures visible with actionable guidance?

SB-09-Web



SB-09-Mobile



SB-10 Security, Compliance, and Enterprise Controls

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Data protection and compliance readiness
- Enterprise support workflows and accountable admin actions

Technical requirement focus:

- Auth enforcement, role checks, rate limiting
- PII handling, retention policy operations, auditability
- Support console and organization membership controls

Discussion prompts:

- Which admin operations need stronger confirmation UX?
- Is the compliance request flow understandable for end users?

SB-10-Web



Ads are disabled for this environment

Profile

Signed in as **demo@vehiclevitals.com**

Account Overview

Email
demo@vehiclevitals.com

User ID
gpkqJsGubxfvzT31ohwHnk8D9PK2

Linked providers
Email/Password

[Link Google](#)

[Link Apple](#)

Alerts
Enabled

Push notifications
Disabled

Reminder lead time
14 days

Nearby services
0 results

Garage makes
3

[Sign out](#)

Maintenance Alert Preferences

Control whether maintenance reminders are generated and when they should appear before service is due.

Upcoming Tasks uses these settings to decide which services appear now and how far in advance to surface them. Higher values show work earlier; lower values keep the queue focused on near-term needs.

Enable maintenance alerts ☒

Preferred reminder lead time (days)

Example: 14 means the app starts showing work about two weeks before it estimates the service will be due.

14

Average driving distance (miles/day)

This helps convert mileage-based schedules into a practical reminder window. If you usually drive 35 miles a day, a 14-day lead time shows tasks roughly 490 miles before they are due.

35

[Save Alert Preferences](#)

Home Address

Save your home address so we can surface nearby repair shops and dealerships.

Street address

123 Main St

Apt, suite, unit (optional)

Apt 4B

City

Nashville

State / province

TN

ZIP / postal code

37203

Search radius (miles)

25

Service type

All providers

Use my vehicle make for dealership matching ☒

BMW

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SB-10-Mobile

Authenticated Profile
Screen



Account Overview



Maintenance Alert
Preferences



Home Address and Provider
Radius



Save Preferences and Re-run Provider Lookup

SB-11 Release Confidence Gates (R1) and Beta Evidence

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Move from implemented to production-credible
- Provide evidence beta testers can challenge and trust

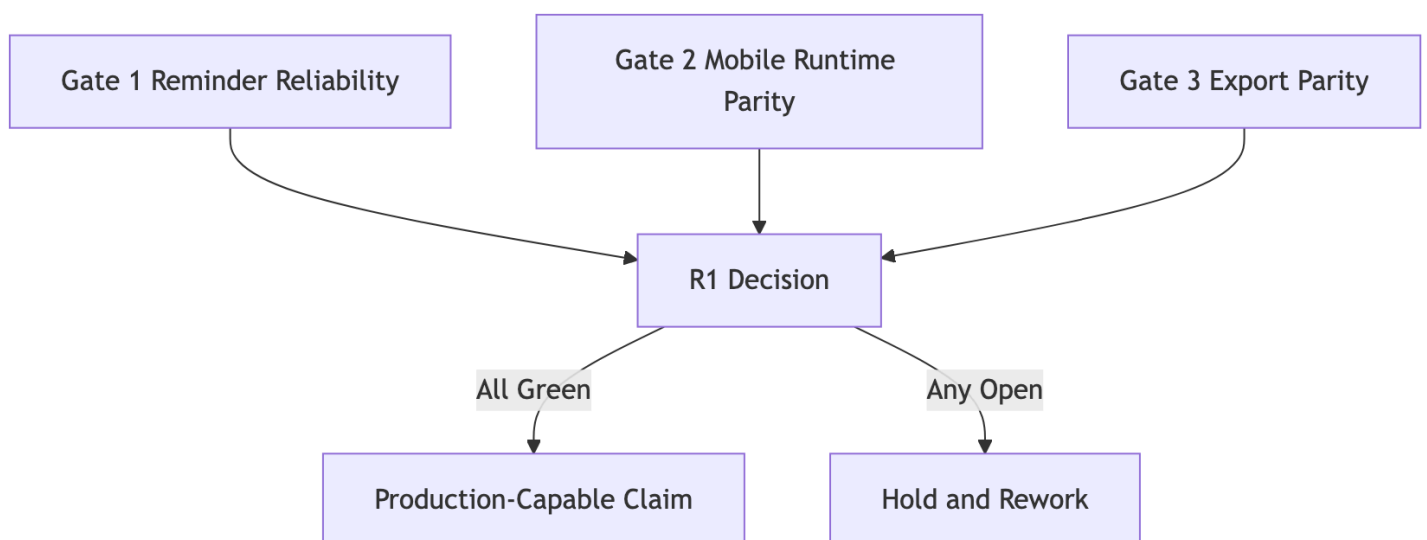
Technical requirement focus:

- Gate 1 reminder reliability
- Gate 2 mobile runtime parity validation
- Gate 3 export parity signoff

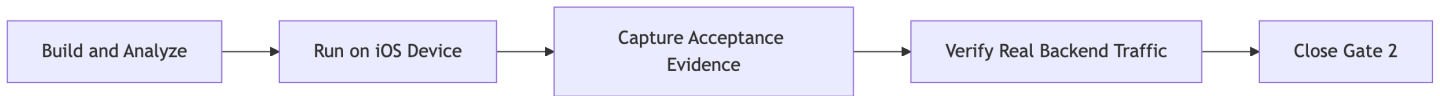
Discussion prompts:

- Which gate evidence is convincing for go/no-go?
- What additional user-observed evidence is needed?

SB-11-Web



SB-11-Mobile



SB-12 Roadmap Rework Workshop Board

Links on this page: [Web Visual](#), [Mobile Visual](#)

Business requirement focus:

- Prioritize what increases retention and revenue fastest
- Align roadmap with scope matrix and resource reality

Technical requirement focus:

- Sequence dependencies and risk
- Keep architecture and security constraints explicit in roadmap decisions

Discussion prompts:

- Which rework items are mandatory before scale?
- What can move to later phases without strategic damage?

SB-12-Web



SB-12-Mobile

Parity and Stability



UX Refinement



Advanced Intelligence
Features



Fleet and Ecosystem
Expansion

Use this during beta sessions.

Storyboard	Business Clarity (1-5)	Technical Feasibility (1-5)	Web/Mobile Consistency (1-5)	Rework Priority (H/M/L)	Notes
SB-01					
SB-02					
SB-03					
SB-04					
SB-05					
SB-06					
SB-07					
SB-08					
SB-09					
SB-10					
SB-11					
SB-12					

6) Decision Log Stub

Date	Storyboard	Decision	Owner	ETA	Validation Method